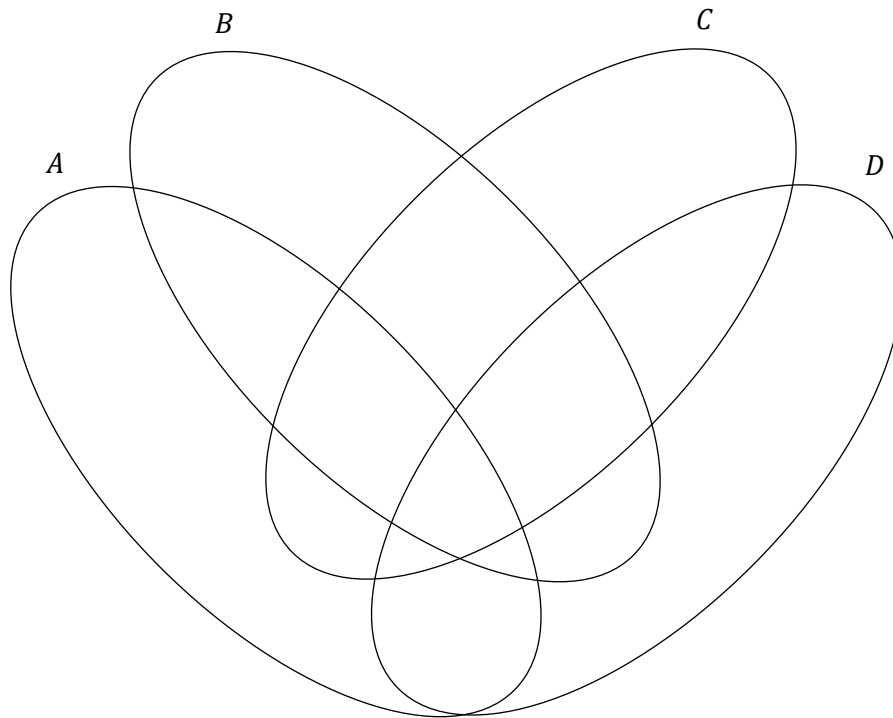


1. There are 27 people in a café. 22 people have a drink in front of them, and 16 people are eating. If 12 people are eating with a drink in front of them, then how many people are neither eating or drinking?
2. Mark has a pack of cards numbered consecutively from 0 to 1000. How many of his cards contain a multiple of 2 or 3?
3. The School currently contains 760 pupils, of which 197 are the in Remove. Given that exactly 600 pupils attended Abbey, what can you say about the number of Remove pupils that attended Abbey?
4. 34 people in the village of Little Hamlets have red cars, and 25 have blue cars. What are the largest and smallest possible number of people who have either a red car or a blue car? Draw Venn diagrams to show both extremes.
5. In a particular school, every pupil must take at least one of Music or Drama at GCSE. A quarter of the pupils who take Music also take Drama, and a third of the pupils who take Drama also take Music. What proportion of pupils at the school take both Music and Drama.
6. A certain school offers GCSEs in French, German, and Russian. There are 120 pupils in the year-group, and the school rules insist that each pupil studies at least one of the above languages. Overall there are 80 pupils studying French, 50 studying German, and 37 studying Russian. There are 28 pupils that study both French and German, but not Russian, and 8 pupils that study Russian and German, but not French. There are 2 pupils who study all three languages.
 - a. How many pupils study German, but not French or Russian?
 - b. How many pupils study French and Russian, but not German?
7. Fifty university science students have to choose exactly two from geology, computing, and oceanography. Forty choose geology and and thirty-five choose computing.
 - a. How many choose oceanography?
 - b. How many do both geology and computing?
8. How many numbers between 1 and 5000 inclusive are multiples of 3, 5, or 7?
9. How many prime numbers are less than 120?
[Try to use the inclusion-exclusion principle for this: don't just list the primes!].
10. A hundred people were surveyed as to what they ate for breakfast this morning. 93 said that they ate cereal; 87 said that they ate toast; 68 said that they ate eggs; and 59 said that they ate bacon. What is the minimum number of people asked who ate all four of these things?
11. In a collection of 72 snails:
 - 35 have bars on their shells;
 - 30 have spots on their shells;
 - 25 have whorls on their shells;
 - 13 have bars and spots on their shells;
 - 11 have spots and whorls on their shells;
 - 7 have bars and whorls on their shells;
 - 8 have neither bars nor spots nor whorls on their shells.

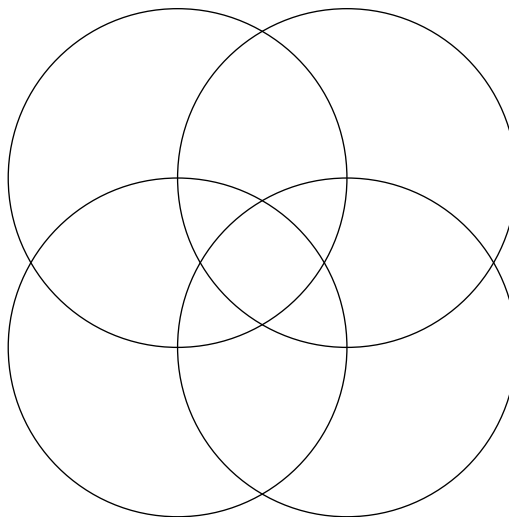
How many snails have bars and spots and whorls on their shells?

12. Use the inclusion-exclusion principle for two sets to find an expression for $n(A \cup B \cup C)$.

13. Guess (if you haven't already been told) what the inclusion-exclusion principle looks like in the case of four sets. Use a 'tally' or otherwise to check that your formula is correct. You can use the diagram below to help you.



14. Why does the diagram above look so complicated? Why **doesn't** the diagram below suffice as a Venn diagram for four sets?



15. [Investigate] What might Venn diagrams for higher number of sets look like? Can you find a systematic way to draw a Venn diagram for n sets? Is there a maximum number of sets that can be represented on a Venn diagram?